

Ollama

Ollama

- 1. Large Language Models (LLM)
- 2. Ollama Installation
 - Script Installation
- 3. Using Ollama
- 4. Uninstalling Ollama
- References

Demonstration Environment

Development Board: RDK X5 mainboard

SD Card: 64G

Tutorial Scope

Mainboard Model	Supported
rdk x5 4gb	√
rdk x5 8gb	√

Ollama is an open-source tool designed to simplify the deployment and running process of large language models, allowing users to use high-quality language models in a local environment.

1. Large Language Models (LLM)

Large Language Models (LLMs) are a class of advanced text generation systems based on artificial intelligence technology. Their main feature is their ability to learn and understand human language through large-scale training data and to generate naturally fluent text.

2. Ollama Installation

Note: It is recommended to use the image with the environment configured by Yahboom. It takes a long time to configure the environment personally, and there may be exceptions during installation on different system versions.

The tutorial demonstrates how to install Ollama on the rdk x5 mainboard using a script. (**The large language model image provided by Yahboom has the environment configured and does not need to be installed again.**)

Script Installation

```
sudo apt install curl -y
sudo curl -fsSL https://ollama.com/install.sh | sh
```

The entire installation process takes a long time, please wait patiently!

Note: It is normal for ollama pull to fail; you can try multiple times, or pull after setting up a proxy. The method for setting up a proxy requires you to search online yourself.

It is complete when the installation is displayed:

```
root@kali:~# sudo curl -fsSL https://ollama.com/install.sh | sh
>>> Cleaning up old version at /usr/local/lib/ollama
>>> Installing ollama to /usr/local
>>> Downloading Linux arm64 bundle
##### 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/systemd/system/ollama.service.
pcilib: Cannot open /proc/bus/pci
lspci: Cannot find any working access method.
pcilib: Cannot open /proc/bus/pci
lspci: Cannot find any working access method.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
```

3. Using Ollama

Enter `ollama` in the terminal to see the prompt:

```
root@kali:~# ollama
Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve      Start ollama
  create     Create a model
  show       Show information for a model
  run        Run a model
  stop       Stop a running model
  pull       Pull a model from a registry
  push       Push a model to a registry
  signin     Sign in to ollama.com
  signout    Sign out from ollama.com
  list       List models
  ps         List running models
  cp         Copy a model
  rm         Remove a model
  help       Help about any command

Flags:
  -h, --help      help for ollama
  -v, --version    Show version information

Use "ollama [command] --help" for more information about a command.
```

Command	Function
ollama serve	Start ollama
ollama create	Create a model from a model file
ollama show	Show model information
ollama run	Run a model
ollama stop	Stop a model
ollama pull	Pull a model from the registry
ollama push	Push a model to the registry
ollama list	List models

Command	Function
ollama ps	List running models
ollama cp	Copy a model
ollama rm	Remove a model
ollama help	Get help information about any command

4. Uninstalling Ollama

- Delete the service

```
sudo systemctl stop ollama
sudo systemctl disable ollama
sudo rm /etc/systemd/system/ollama.service
```

- Delete the file

```
sudo rm $(which ollama)
```

- Delete the model and service user and group

```
sudo rm -r /usr/share/ollama
sudo userdel ollama
sudo groupdel ollama
```

References

Ollama

Official Website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>